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NOTE AND RECORD

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Loss of tail sheath (false autotomy) in South African Cape spiny mice (*Acomys subspinosus*)

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1 | INTRODUCTION

Tail autotomy is known in different species of reptiles, amphibians and rodents (Arnold, 1984; Layne, 1972; Shargal, Rath-Wolfson, Kronfeld, & Dayan, 1999) and can either entail the simultaneous loss of an entire tail or part thereof through breakage across the vertebra (true autotomy) or the exclusive loss of the tail sheath whereupon the denuded part disappears later (false autotomy) (Dubost & Gasc, 1987; Mohr, 1941). False autotomy has been reported in several rodent taxa totalling to 32 species in 20 genera and five families worldwide, obviously having evolved independently several times (Shargal et al., 1999; Seifert et al., 2012; this study). One example is the genus *Acomys* I. Geoffroy Saint-Hilaire, 1838 with two species from Israel and three species from Kenya which are known to lose the tail skin (Seifert et al., 2012; Shargal et al., 1999).

Here, we use an experimental approach to test a further species for false autotomy. We test whether tail sheath loss in South African *A. subspinosus* is triggered directly by grabbing and holding the tail, similar to a potential predator.

2 | MATERIALS AND METHODS

Live-trapping and observations were carried out on the Matsikamma mountain (south of Vanrhynsdorp, Western Cape, South Africa; Figure 1a) on the farm Sewefontein (S31°44.211′, E18°49.496′, elevation 660 m) and Nawelskloof (S31°44.715′, E18°49.470′, 715 m). Over 19 nights (2.–22.10.2016), 20–70 mammal traps, baited with a mixture of peanut butter and rolled oats, were laid out. Captured animals were removed, identified and sexed. Two *A. subspinosus* that lost their tail sheath during capture were kept in captivity (in terraria with food and water) and observed for several days.

Four *A. subspinosus* individuals (two females, one male and one individual of unknown sex) were held gently at the base of their tail with applying only slight pressure and without pulling. As the mice did not exhibit an escape response, we attempted to elicit such a response by approaching them with a cloth. The behaviour of the mice was recorded, including the loss of tail sheath, possible nibbling or licking at the tail, locomotion and any potentially noticeable behaviour different to normal.

3 | RESULTS AND DISCUSSION

During several years of live-trapping small mammals on the Gifberg near Vanrhynsdorp (South Africa), some of the individuals captured were found lacking a part of the tail. However, only in Cape spiny mice, *Acomys subspinosus* (Waterhouse, 1838) were individuals found to occasionally miss the entire tail (alone in 2016, one male and one juvenile of 14 caught, and in 2017, three of 14 caught). While in other years, we limited touching of animals to a minimum; in 2016, the animals were handled by grasping the animals' body or base of the tail and fixing their head (V grip). When releasing the animals, one female *A. subspinosus* caught our attention because it lost the sheath of its tail. Another (juvenile) *A. subspinosus* lost its tail skin during restraint. During the course of this study, we never observed this in other species caught such as Namaqua rock mice, *Micaelamys namaquensis* (A. Smith, 1834); Verreaux's mice, *Myomyscus verreauxi* (A. Smith, 1834); Four-striped grass mice, *Rhabdomys pumilio* (Sparrman, 1784); and Cape Rock elephant-shrews, *Elephantulus edwardii* (A. Smith, 1839) (P. Wester, unpubl. data).

However, after our anecdotal observation, it remained unclear whether the observed skin loss happened accidently, for example, by the tail being caught in the trap door or was directly linked to handling (e.g. pulling tail that was held) and what the mechanism was.

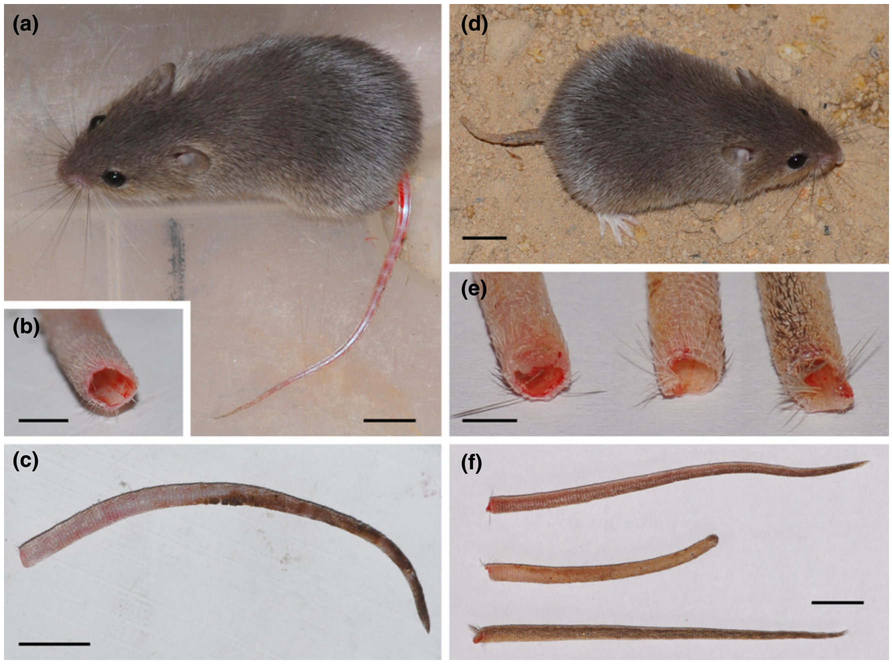

FIGURE 1 *Acomys subspinosus* individual with denuded part of tail (a) just after the loss of tail skin: basal end (b, note limited bleeding) and complete tail sheath (c), captive *A. subspinosus* with older denuded part of the tail already desiccated and shortened (d); experimentally separated tail sheaths: basal end (e) and complete (f): note the longer fur at the base of the sheath. Scale bars = 1 cm except b and e = 0.25 cm [Colour figure can be viewed at wileyonlinelibrary.com]

The experiments showed that all of the *A. subspinosus* tested lost their tail sheath when escaping. The tail sheath separated easily from the subjacent muscles and vertebrae at the base of the tail (Figure 1).

This study shows that false autotomy of tail skin occurs in *A. subspinosus*. In *Acomys* species as well as in further rodent taxa, tail sheath loss is facilitated by weak zones in the dermis and the loose attachment of the tail skin to the underlying structures compared to nonautotomy species (Cuenot, 1907; Layne, 1972; Shargal et al., 1999). Furthermore, these predetermined breaking points result in limited bleeding (Juškaitis, 2008; Shargal et al., 1999), a phenomenon we also observed in *A. subspinosus* (Figure 1a,b,e).

In the *A. subspinosus* individuals kept in captivity, the denuded part of the tail dried up as reported in previous studies (Layne, 1972; Michener, 1976; Shargal et al., 1999). One *A. subspinosus* individual was found without tail the next morning, whereas the other mice kept its stub for several days (Figure 1d). The stub length decreased over time until it was lost entirely. Similarly, other studies report chewing off or eventually shedding the stub (Layne, 1972; Michener, 1976; Shargal et al., 1999). We observed chewing or licking of the tail stub at least five times in one of the captive animals. However, as no detailed observations were conducted, we were unable to determine whether consumptions or/and shedding were responsible for stub loss. For *Acomys*, rapid healing of torn skin has been reported (Seifert et al., 2012; see also Michener, 1976 for rapid healing in rats) and several studies found no disadvantage of tail loss (i.e. locomotory function, life expectancy; Shargal et al., 1999). None of the tailless *A. subspinosus* in this study exhibited apparent impairments of running and climbing abilities. However, as we did not quantify these variables and only conducted short-term observations, we cannot exclude negative effects of tail loss on locomotion and/or survival in *A. subspinosus*.

Similar to true autotomy, false autotomy has been interpreted as a predator escape mechanism in other rodents (Layne, 1972; Michener, 1976; Shargal et al., 1999 and citations therein). If such tail loss has indeed survival benefits when attacked by a predator, this could account for the large proportion of tailless individuals in wild populations of *Acomys* spp. (see also Seifert et al., 2012; Shargal et al., 1999).

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REFERENCES

Arnold, E. N. (1984). Evolutionary aspects of tail shedding in lizards and their relatives. *Journal of Natural History*, *18*, 127–169. https://doi.org/10.1080/00222938400770131

Cuenot, M. L. (1907). L'autotomie caudale chez quelques mammiferes du groupe des rongeurs. *Comptes rendus des séances de la Société de biologie et de ses filiales*, *62*, 174–176.

Dubost, G., & Gasc, J.-P. (1987). The process of total tail autotomy in the South-American rodent, Proechimys. *Journal of Zoology*, *212*, 563–572. https://doi.org/10.1111/j.1469-7998.1987.tb02924.x

Juškaitis, R. (2008). *The Common Dormouse Muscardinus avellanarius: Ecology, Population Structure and Dynamics*. Vilnius, Lithuania: Institute of Ecology of Vilnius University Publishers.

Layne, J. N. (1972). Tail autotomy in the Florida Mouse, *Peromyscus floridanus*. *Journal of Mammalogy*, *53*, 62–71. https://doi.org/10.2307/1378827

Michener, G. R. (1976). Tail autotomy as an escape mechanism in *Rattus rattus*. *Journal of Mammalogy*, *57*, 600–603. https://doi.org/10.2307/1379315

Mohr, E. (1941). Schwanzverlust und Schwanzregeneration bei Nagetieren. *Zoologischer Anzeiger*, *135*, 49–65.

Seifert, A. W., Kiama, S. G., Seifert, M. G., Goheen, J. R., Palmer, T. M., & Maden, M. (2012). Skin shedding and tissue regeneration in African spiny mice (*Acomys*). *Nature*, *489*, 561–566. https://doi.org/10.1038/nature11499

Shargal, E., Rath-Wolfson, L., Kronfeld, N., & Dayan, T. (1999). Ecological and histological aspects of tail loss in spiny mice (Rodentia: Muridae, *Acomys*) with a review of its occurrence in rodents. *Journal of Zoology*, *249*, 187–193. https://doi.org/10.1111/j.1469-7998.1999.tb00757.x

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